CSIRO

The Commonwealth Scientific and Industrial Organisation (CSIRO) Giant Oystercracker Dart (Trachinotus anak) fish project represents a welfare-focused development for Australia’s aquaculture industry.

This initiative demonstrates how welfare considerations can guide which fish species and farming methods work well for Australia, while helping to grow Australia's domestic white-fish industry (CSIRO 2024).

Welfare-first species selection

Dr Polly Hilder's team adopted a ‘holistic approach incorporating sustainability and welfare’ from the beginning of the project.

The team established farming practices that prioritised animal wellbeing while supporting commercial success. The selection of the Giant Oystercracker reflects this welfare-first philosophy, as the natural characteristics of this species reduce common welfare challenges in aquaculture.

Unlike many farmed fish species, Giant Oystercrackers are naturally cooperative and do not display cannibalism at weaning, or aggression throughout their production cycle.

These behaviours can create challenges in aquaculture and impact welfare outcomes. The non-aggressive nature of the Giant Oystercracker means stressful management practices used to control aggressive behaviour are not required. This results in both welfare and economic benefits.

The Giant Oystercracker is native to tropical waters from Exmouth in Western Australia, across the Northern Territory and Queensland and as far south as Coffs Harbour in New South Wales. Because they are already adapted to local conditions, there is less need for intensive environmental controls and artificial habitat modifications that can compromise welfare in other aquaculture systems.

Welfare-driven production systems

CSIRO's approach goes beyond choosing the right species. It also involves designing the entire production system around welfare principles.

By developing management protocols that work with natural behaviours, CSIRO is creating sustainable farming practices that prioritise fish wellbeing while maintaining commercial competitiveness.

For example, the CSIRO Giant Oystercracker fish project uses proactive health care plans. These include regular checks on fish health, water quality, and the overall environment to identify and address potential problems early.

The cooperative nature of the Giant Oystercracker also allows for bigger fish populations to be raised together without harming individual fish welfare.

Together, these strategies help maintain fish wellbeing and support strong, disease-resistant populations, leading to more sustainable aquaculture.

Industry transformation potential

This change in approach demonstrates how welfare can transform the way aquaculture is developed.

According to CSIRO's National Protein Roadmap, strengthening and diversifying white-flesh fish production in Australia through the development of a new species will increase consumer choice and support the production of nutritious, locally produced seafood. It will also help create jobs and utilise local resources.

This demonstrates that prioritising animal welfare can deliver significant economic and social benefits, creating a strong foundation for industry growth that benefits both fish and communities.

Broader significance

The CSIRO Giant Oystercracker project shows how scientific innovation can balance the goals of commercial aquaculture with improved animal welfare.

Rather than being a barrier to productivity, CSIRO has demonstrated that choosing species with welfare in mind can improve both ethical and economic outcomes.

Learn more about CSIRO’s research and innovation in aquaculture.

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Video: Giant Oystercracker fish (without sound)